Five Top Viruses and Worms of All Time

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Abstract

In the growing world of technology, computer viruses and worms are constant dangers to the security of digital systems. From the beginning of the internet to modern-day cybersecurity practices, various malicious programs have tried to exploit many weaknesses in software, networks, and even human behavior. Every major hacking incident has highlighted areas for improvement, particularly in detection, prevention, and response time. This paper will discuss five of the most devastating computer worms of all time: the Morris Worm, ILOVEYOU Worm, Nimda Worm, Code Red Worm, and the Melissa Virus. We will focus on the causes, consequences, and their lasting impact.

The Morris Worm was released onto the Internet on November 2, 1988, by Robert Tappan Morris, a 23-year-old Cornell University graduate. He initially created the worm as a harmless experiment to test how far computer worms could navigate through the Internet. However, there was a coding flaw within the program that caused it to replicate itself without needing a software host. It infected around 6,000 of the approximately 60,000 online computers at that time. During the hack, the Morris Worm exploited Unix systems using brute force attacks, sendmail, and finger services.

The overall damages ranged from 100,000 to the millions, with additional damages not even being accounted for. This hacking incident led to the creation of the CFAA, or Computer Fraud and Abuse Act, which advocated for stronger policies, prevented unauthorized access, and repaired damaged computers. Pushing past just the technical damage, the Morris Worm had a considerable role in bringing about more cybersecurity laws/policies. Especially sparking widespread discussion about safe and ethical programming, and the importance of risks. Finally, it made governments and organizations realize the importance of stronger defenses, monitoring systems, and communication.

The ILOVEYOU Worm was released onto the Internet on May 4, 2000, by Onel de Guzman, a 24-year-old student at AMA Computer College in the Philippines. He initially created the worm as a way to access the internet by stealing passwords. In this instance, the virus disguised itself as a love letter with the subject “ILOVEYOU” and an attached file named “LOVE-LETTER-FOR-YOU.TXT”. It exploited a vulnerability within Windows where the file was titled as a .txt, which stands for text file. In contrast, it was really a VBScript file, which stands for Visual Basic Script.

A VBScript file contained code that, when interacted with, could process functions. In this instance, when a user opened the attached file, it ran a Visual Basic script that began to steal passwords, overwrite files, and replicate across multiple computers, similarly to the Morris worm. Its impact was the same, with over 45 million computers and their emails being infected in just 24 hours. The damage caused by the ILOVEYOU worm was estimated to be around three to fifteen billion dollars, making it one of the most expensive cyberworm attacks ever recorded. Besides the cost in damages, this attack showcased how easily humans could be manipulated by something as simple as a love letter.

The virus pushed the government to adopt stronger cybersecurity policies, improve email filtering, and urge the public to be more aware of cybersecurity threats. Even in today's age, phishing and malware are still widely used as foundational tactics by every scammer. The ILOVEYOU virus taught us not to trust everything we see on the internet. The virus served as a strategy, ranging from the usual coping hacks to simple human error and emotion.

The Nimda Worm was released onto the Internet on September 18, 2001, and it was one of the fastest-spreading viruses at that time. It was created to cause traffic slowdowns across the Internet. In this instance, the virus spread through 4 different methods, which made it very versatile and hard to stop. The first method was through email, similar to the ILOVEYOU virus, where infected attachments were sent to people within the victim's contact list. The second method involved network shares containing folders across shared networks and computers, enabling the virus to infect and spread through these folders.

The third method was through vulnerable IISs or Internet Information Servers. These servers contained sensitive information regarding the authentication of users. The fourth method involved exploiting backdoors left behind by previous hacks, including the Code Red Worm and

other major malware attacks. These methods made the Nimda Worm one of the most influential and dangerous viruses of its time. The significant internet slowdowns to all business and government facilities worldwide called for better network protection. It introduced the idea of keeping system vulnerabilities patched, restricting access to shared networks, and improving detection and monitoring for vulnerabilities, such as the backdoor method. All of which are still used to this day.

The Code Red Worm was released onto the Internet on July 13, 2001, and it became one of the most dangerous worms to date. Its tactic was to target Windows systems using Microsoft Internet Information Services (IIS) web servers while exploiting a vulnerability. The vulnerability, named MS01-33, was identified as a buffer overflow error in the idq.dll library. It was also reported that the vulnerability was easy to exploit just by sending the server a long request, such as spamming a bunch of letters, numbers, and symbols. After the server was infected, it was programmed to send a message: “Hacked by Chinese.”

The next step was to infect other online domains by sending a Denial of Service (DoS) attack. It even attempted to send a DOS attack to the White House web server, which at the time showcased just how dangerous the worm was. Within hours, the virus had allegedly hacked over 300,000 servers and slowed down internet traffic. This virus could have been prevented because, earlier that day, a security patch was implemented to address a vulnerability, but many users had not updated it. The entire incident made people realize that routine software patches and updates should be implemented to ensure the safety of their systems, a practice that remains in place today.

The Melissa Virus was released onto the Internet in March 1999 by David Lee Smith. The name was reportedly taken from a stripper that Smith had met while in Florida. It was

designed to take over users' Microsoft Outlook accounts using a similar tactic to the ILOVEYOU virus. The tactic was to send an attachment that would intrigue the user. If the user clicked on the attachment, it would hijack Outlook and turn off security features in Word 97 and Word 2000, which protected the user from unauthorized macros.

The macro would then automatically send itself to the first 50 people in the victim's address book, which flooded email servers. The virus itself did not destroy files or damage systems but rather caused massive disruption across organizations, governments, and larger mail servers. The virus had caused email servers to become so flooded that even Microsoft itself had to shut down its systems to prevent further damage. The damages at the time were considered high, with an estimated cost of $80 million for repairing and cleaning all systems. The Melissa virus demonstrated how human emotion and curiosity could be exploited to spread malware. It also promoted stronger email filtering and user awareness to prevent such social-engineering attacks in the future.

While these cybercrimes were major events, they did not go unpunished for their creators, with some remaining unresolved. Robert Tappan Morris was identified as the creator of the Morris Worm. He became the first person ever convicted under the Computer Fraud and Abuse Act (CFAA), a major law in cybercrime cases. Onel de Guzman was suspected of creating the ILOVEYOU Worm, but at that time in the Philippines, there were no laws addressing cybercrime. The creator of the Melissa Virus, David Lee Smith, was sentenced to 20 months in federal prison and fined $5,000. However, the creators of the Nimda and Code Red Worms were never found because tracing digital footprints and catching cybercriminals was difficult back in the day.

Throughout history, computer viruses and worms have wreaked havoc across the cyber world. The Morris Worm, ILOVEYOU Worm, Nimda Worm, Code Red Worm, and Melissa Virus have demonstrated how systems are continually tested and improved to withstand these attacks. Each of these worms/viruses has exploited many different weaknesses in history. The Morris Worm exploited weak password vulnerabilities within Unix systems, leading to the creation of new computer crime laws such as the Computer Fraud and Abuse Act (CFAA). The ILOVEYOU Worm exploited human curiosity and emotion through emails and attachments, demonstrating how social engineering was becoming a hacking method.

The Nimda Worm demonstrated how multiple attack methods could be combined into a single effective strategy, proving that cybersecurity professionals should be prepared for anything and anticipate the possibility of various attacks. The Code Red Worm was a wake-up call for all, demonstrating that even one unchecked vulnerability could lead to massive internet shutdowns and service issues worldwide. The Melissa Virus showcased how easily human emotion and curiosity could be manipulated. With just a simple email, it was able to bring down multiple company systems and cost millions in damages. Each of these viruses and worms didn't just change the way we saw the internet, but also changed the way we protect our systems.

Works Cited

Gillis, A. S. (2021). What is the Melissa virus? Retrieved from https://www.techtarget.com/searchsecurity/definition/Melissa-virus

Government, F. (2019). The melissa virus | Federal Bureau of Investigation. Retrieved from https://www.fbi.gov/news/stories/melissa-virus-20th-anniversary-032519

Hanna, K. T. (2021). What is the Nimda computer virus? definition from searchsecurity. Retrieved from https://www.techtarget.com/searchsecurity/definition/Nimda

James, H. J. (2023). Iloveyou virus attacks computers: Research starters: EBSCO research. Retrieved from https://www.ebsco.com/research-starters/computer-science/iloveyou-virus-attacks-computers

Root, E. (2022). The evolution of security: The story of code red. Retrieved from https://www.kaspersky.com/blog/history-lessons-code-red/45082/

SecPoint. (n.d.). Top 10 worms - most famous internet worms security. Retrieved from https://www.secpoint.com/top-10-worms.html

Spivack, E. (2025). “iloveyou”: How the infamous Computer Worm wreaked havoc. Retrieved from https://www.history.com/articles/i-love-you-computer-worm

The morris worm, Hilarie Orman purple streak. (n.d.). Retrieved from https://www.cs.umd.edu/class/fall2023/cmsc614/papers/morris-worm.pdf